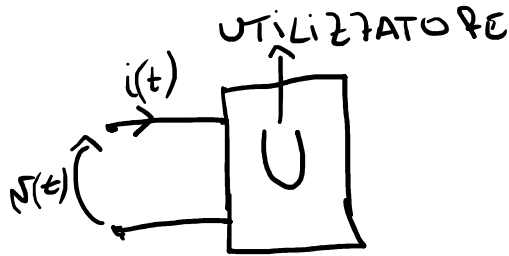


POTENZA IN REGIME SINUSOIDALE

mercoledì 4 maggio 2022 16:29



$$P(t) = v(t) \cdot i(t)$$

POT. INSTANT.
ASSORBITA
DA U

$$v(t) = \hat{V} \cos(\omega t) \quad \alpha_v = 0$$

$$i(t) = \hat{I} \cos(\omega t + \alpha_i) = \hat{I} \cos(\omega t - \varphi)$$

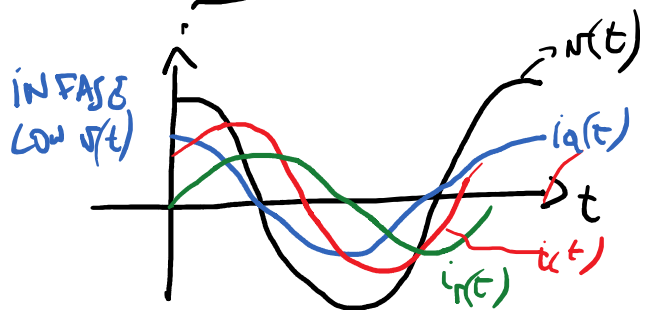
Scomponiamo la corrente $i(t)$ in due componenti:
una in fase con $v(t)$ ed una in quadratura:

$$i(t) = i_a(t) + i_r(t)$$

$$i(t) = \hat{I} \cos(\omega t - \varphi) \underset{\uparrow}{=} \hat{I} [\cos \varphi \cos(\omega t) + \sin \varphi \sin(\omega t)]$$

$$[\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta]$$

$$\begin{cases} i_a(t) = \hat{I} \cos \varphi \cos(\omega t) \\ i_r(t) = \hat{I} \sin \varphi \sin(\omega t) \end{cases}$$



$$P(t) = v(t) i(t) = v(t) (i_a(t) + i_r(t)) = \underbrace{v(t) i_a(t)}_{P_a(t)} + \underbrace{v(t) i_r(t)}_{P_r(t)}$$

POT. ATTIVA Istantanea POT. REATTIVA Istantanea

$$P_a(t) = \hat{V} \cos(\omega t) \cdot \hat{I} \cos \varphi \cos(\omega t) = \hat{V} \hat{I} \cos \varphi \cos^2(\omega t)$$

$$P_a(t) = \hat{V} \cos(\omega t) \cdot \hat{I} \cos \varphi \cos(\omega t) = \hat{V} \hat{I} \cos \varphi \cos^2(\omega t)$$

$$P_r(t) = \hat{V} \cos(\omega t) \cdot \hat{I} \sin \varphi \sin(\omega t) = \hat{V} \hat{I} \sin \varphi \cos(\omega t) \sin(\omega t) \quad \sim \text{VALORI DI PICCO}$$

$$\hat{V} = \sqrt{2} \cdot V$$

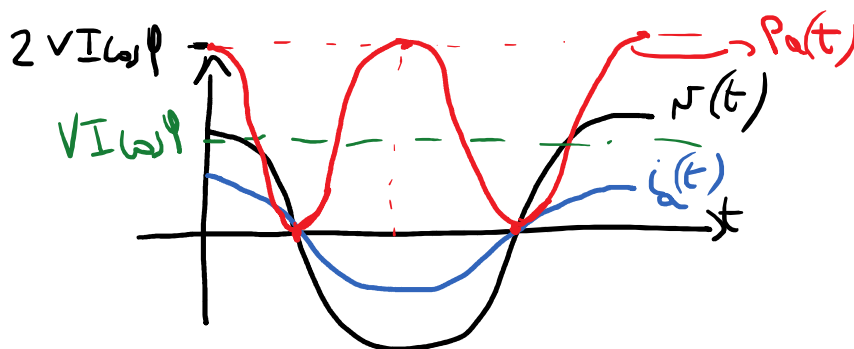
$$[\sin(2\alpha) = 2 \sin \alpha \cos \alpha]$$

$$\hat{I} = \sqrt{2} \cdot I$$

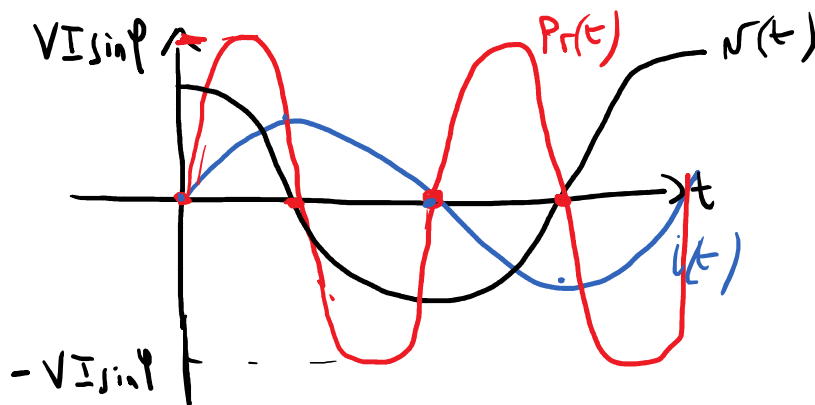
$$P_r(t) = \frac{\hat{V} \hat{I}}{2} \sin \varphi \sin(2\omega t)$$

$$\begin{cases} P_a(t) = 2 V I \cos \varphi \cos^2(\omega t) \\ P_r(t) = V I \sin \varphi \sin(2\omega t) \end{cases} \quad \sim \text{VAL. EFFICACE}$$

$$-\frac{\pi}{2} < \varphi < \frac{\pi}{2}$$



$$P_r(t) = V I \sin \varphi \sin(2\omega t)$$



È ASSOCIATA
ALL'ENERGIA UTILE

$$\begin{cases} P_a(t) \geq 0 \text{ sempre} \Rightarrow \text{è una potenza che va sempre dal gener. verso il carico} \\ P_r(t) \text{ sia positiva che negativa} \Rightarrow \text{è associata ad una potenza che può andare dal g. al carico o viceversa} \end{cases}$$

$$\begin{cases} P_a(t) \Rightarrow \text{associata al resistore} \\ P_r(t) \Rightarrow \text{condensatore e induttore} \end{cases}$$

L'energia assorbita dal carico U:

$$\begin{aligned} \int_0^T p(t) dt &= \int_0^T P_a(t) + P_r(t) dt = \\ &= \int_0^T 2VI \cos \phi \cos^2(\omega t) dt + \int_0^T VI \sin \phi \sin(2\omega t) dt \\ &\quad \downarrow \qquad \qquad \qquad \downarrow \\ 2VI \cos \phi \int_0^T \cos^2(\omega t) dt &= VI \sin \phi \int_0^T \sin(2\omega t) dt \\ &\quad \left[\cos^2 x = \frac{1 + \cos(2x)}{2} \right] \qquad \qquad \qquad = 0 \\ &= 2VI \cos \phi \int_0^T \frac{1 + \cos(2\omega t)}{2} dt \quad \left| \begin{array}{l} \text{il bilancio energetico} \\ \text{è nullo} \end{array} \right. \\ &= VI \cos \phi \cdot T \end{aligned}$$

LAVORO UTILE

$$P_Q = \frac{1}{T} \int_0^T p_Q(t) dt = \frac{1}{T} V I \cos \varphi T$$

$$P_Q = V I \cos \varphi$$

FATTORE DI
POTENZA

POT. MEDIA
ASSORBITA
DAL CARICO $\Rightarrow$ LAVORO
UTILE

POTENZA COMPLESSA

mercoledì 4 maggio 2022 17:21

$$\underline{N} = \underline{V} \underline{I}^*$$

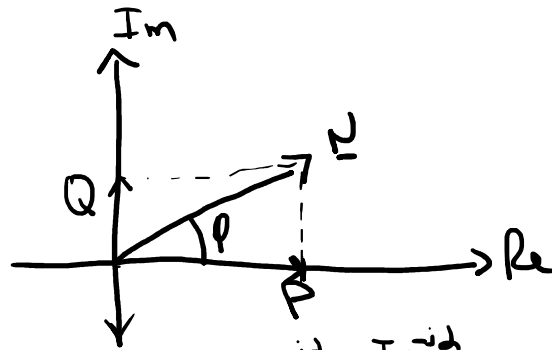
$$\underline{X} = X_r + j X_I = X e^{j\alpha}$$

$$\underline{X}^* = X_r - j X_I = X e^{-j\alpha}$$

$$= V e^{j\alpha_V} \cdot I e^{-j\alpha_I} = VI e^{j(\alpha_V - \alpha_I)} = VI e^{j\varphi} =$$

$$= \underbrace{VI \cos \varphi}_P + j \underbrace{VI \sin \varphi}_Q = P + jQ$$

POT. REATTIVA



P: POT. ATTIVA [W]

Q: POT. REATTIVA [VAR]

↙
Valori Ampere Reattivi

$$\underline{N} = \underline{V} \cdot \underline{I}^* = \underline{Z} \underline{I} \cdot \underline{I}^* = \underline{Z} \underline{I}^2 = (R + jX) \underline{I}^2$$

$$= \underbrace{R \underline{I}^2}_P + j \underbrace{X \underline{I}^2}_Q$$

$P > 0 \rightarrow$ associata ad R

$Q \rightarrow$ associata a L e C

$$\begin{cases} \rightarrow Q_L = \omega L I^2 > 0 \\ \rightarrow Q_C = -\frac{1}{\omega C} I^2 < 0 \end{cases}$$

In sintesi :

$$P = VI \cos \varphi$$

$$Q = VI \sin \varphi$$

$$N = \underline{V} \underline{I}^*$$

$$|N|$$

Pot. attiva [W]

Pot. reattiva [VAR] VOLTI AMPERI REATTIVI

Pot. complessa [VA]

Pot. apparente [VA]